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N + - Network Fundamentals & Building Networks.

Static Routing Concepts.

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TASK 1.

❖ Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.

- **Network Setup:**
  Two routers were added in Cisco Packet Tracer and connected using a serial DCE/DTE cable.

IP Address Configuration:

Router Interface	IP Address	Subnet Mask
Router 1 – Serial0/0/0	10.0.0.1	255.255.255.252
Router 2 – Serial0/0/0	10.0.0.2	255.255.255.252

Verification:

Both serial interfaces were configured and enabled using the no shutdown command. The DCE side was assigned a clock rate.

Result: The routers were connected through a serial link and configured with IP addresses in the same subnet.

TASK 2.

❖ On Router1 in Cisco Packet Tracer, configure a static route so it can reach a PC connected to Router2's LAN network. Test connectivity using the ping command from Router1 to the PC.

- **Static Route Configuration and Connectivity Test:**

Configure Static Route:
On Router1, a static route was added to reach the LAN network connected to Router2.

Router1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

Test Connectivity:
The ping command was used from Router1 to test the PC connected to Router2's LAN.

Router1# ping 192.168.2.10

Result: The static route allows Router1 to reach Router2's LAN. Successful ping replies confirm that connectivity is working.

TASK 3.

❖ Simulate a Zomato-style delivery network: imagine Router1 represents an Ahmedabad branch and Router2 represents a Surat branch. Configure static routes so that devices in each city's LAN can communicate with each other through the routers.

- **Zomato Style Delivery Network Using Static Routes:**

Network Setup: Router1 represents the Ahmedabad branch, and Router2 represents the Surat branch. Each router is connected to its own LAN.

IP Addressing and Static Routes:

Device / Interface	IP Address
Ahmedabad LAN (Router1)	192.168.1.1/24
Router1 Serial Interface	10.0.0.1/30
Router2 Serial Interface	10.0.0.2/30
Surat LAN (Router2)	192.168.2.1/24

On **Router1**, configure the route to Surat's LAN**:**

ip route 192.168.2.0 255.255.255.0 10.0.0.2

On **Router2**, configure the route to Ahmedabad's LAN**:**

ip route 192.168.1.0 255.255.255.0 10.0.0.1

Connectivity Test: Use the ping command between PCs in both LANs to verify communication.

Result:
Static routes allow devices in the Ahmedabad and Surat branches to communicate through the routers.

TASK 4.

❖ List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points, Think about how static routes might impact delivery or booking updates in these apps.

- **Advantages:**

Simple to Configure: Static routes are manually set and are easy to manage in small networks. For example, a small Swiggy office can use static routes to connect its computers to the main server.

Less Network Overhead: Static routing does not exchange routing updates automatically, so it uses fewer network resources. This can help a small IRCTC office maintain a simple, stable network connection.

Disadvantages:

No Automatic Updates: If a network path fails, static routes do not automatically find another path. For example, Swiggy delivery updates may be delayed if the server connection fails and the route is not changed manually.

Difficult to Manage Large Networks: Every route must be configured and updated manually. For example, if IRCTC adds many new booking offices, administrators must update routes, which takes time and can cause errors.